INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALTLAKE
(School of University of Engineering and Management, Kolkata)

End Semester Examination, December - 2024

Degree: **B.C.A**, Semester: **Third**

Stream: **BCA**

Paper Code: **BCACC301**

Paper Name: **Data Structures with C**

Full Marks:100 Duration:3 Hours

Part - A
Attempt 10 questions out of 15 questions
Each question carries 2 Marks (2 X 10)

1. Explain the different operations to be performed on data structures. **2**

2. Explain the concept of Big O and W notation. **2**

3. Why Recursion can be an inefficient way to implement a solution? **2**

4. Differentiate between the iteration and recursion. **2**

5. What is an algorithm? **2**

6. Give an example to show the usefulness of an arrays. **2**

7. What are the basic operations performed on a stack? **2**

8. Explain the following terms: **2**

a) Infix expression
b) Polish notation
c) Reverse polish notation

9. What is the relationship between a queue and its underlying array? **2**

10. What is the deque called that allows insertion at only one end but deletions at both the ends of the lists? **2**

11. How do you represent a queue in computer's memory? **2**

12. What are deques? Explain its two variations. **2**

13. What do you mean by 'Front' and 'Rear'? **2**

14. If you are using C language to implement the heterogeneous linked list, what pointer type will you use? **2**

15. What is a binary tree? Mention the properties of a binary tree. **2**

Part - B

Attempt 6 questions out of 9 questions
Each question carries 5 Marks (5 X 6)

16. Write a C program to get a paragraph of text as input. **5**

17. Write a program to find whether the given number is palindrome or not. **5**

18. With the following declaration: **5**

```
int *p, x[10];

char *t, s[100];
```

Explain each of the following expressions. If there is an error, explain why it is an error.

a. 1. x
   2. x + i
   3. * (x + i)
   4. x++ ;

b. 1. P = x ;
2. *p
3. p++ ;
4. p++ ;
5. p - - ;
6. - -p ;

19. Write a function that checks if given integer item is in a list. Traverse the array and check end element of the list. If an element is found return True; if the array is exhausted, return false. **5**

20. Write an algorithm for inserted and deleted in a queue using pointers. **5**

21. Write a pseudocode algorithm for the inorder traversal. **5**

22. When deleting a node, why can't you just use the delete operator to remove it from memory? **5**

23. Write an algorithm for implementing bubble sort, insertion sort, quick sort. **5**

24. Explain how problem of collision can be resolved by separate chaining? **5**

Part - C

Attempt 5 questions out of 8 questions
Each question carries 10 Marks (10 X 5)

25. Write a function that prints, using dereferenced pointers, the elements of an array of type float. **10**

26. Convert X : A + (B * C – (D/E – F) *G)*H into postfix form showing stack status after every step in tabular form. **10**

27. How to declare a structure of a linked list? **10**

28. Write a C program to insert a node at a specific position specified by the user in case of doubly linked list.

29. Write an algorithm to find the inorder successor in threaded binary tree. **10**

30. What are the different types of search techniques you have studied? Explain the one which is more efficient among them. **10**

31. Implement the following function: **10**

```
void marge (int data [ ], size_t n1, size_t n2);
// precondition: The first n1 elements of data are sorted, and the
// next n2 elements of data are sorted (from smallest to largest).
//postcondition: The n1 + n2 elements of data are now completely
// sorted.
```

32. Give the merits of a B-tree index. **10**

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